Define e-Governance. What are benefit of e-Government? Differentiate between traditional government and e-Governance.

# Definition of e-Governance

**e-Governance** (Electronic Governance) refers to the application of Information and Communication Technology (ICT) to deliver government services, exchange information, and perform communication transactions between the government and citizens, businesses, or other government branches.

In simple terms, it is the use of the internet and digital tools to make government processes more efficient, transparent, and accessible to the public.

# Benefits of e-Government

Implementing e-Government systems provides several strategic advantages:

- **Improved Efficiency:** Automation of repetitive tasks and digital workflows speeds up the processing of applications, licenses, and permits.
- **Enhanced Transparency:** Government data and decisions are made available online, allowing citizens to monitor government activities and reducing the "hidden" nature of bureaucracy.
- **Cost Reduction:** In the long run, digitizing records reduces the need for physical paper, massive storage spaces, and a large number of administrative staff.
- **Convenience (24/7 Access):** Citizens can access services (like paying taxes or renewing a bluebook) from their homes at any time, without waiting in long queues.
- **Accountability and Reduced Corruption:** Digital footprints make it easy to track who handled a file and when, which significantly minimizes the opportunities for bribery and manual manipulation.
- **Data-Driven Decision Making:** Real-time data collection helps the government analyze trends and make better policies based on actual facts.


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# Difference Between Traditional Government and e-Governance

| Feature | Traditional Government | e-Governance |
|---|---|---|
| **Primary Medium** | Paper-based, physical files, and manual entry. | Digital-based, databases, and automated systems. |
| **Service Access** | Requires physical presence at a government office. | Can be accessed remotely via web portals or mobile apps. |
| **Operating Hours** | Limited to fixed office hours (e.g., 10 AM to 5 PM). | Services are available 24/7/365. |
| **Speed** | Slow due to manual movement of files and bureaucracy. | Fast due to instant data transfer and automated processing. |
| **Transparency** | High level of secrecy; difficult for citizens to track status. | High level of transparency; real-time tracking is often available. |
| **Citizen Interaction** | Face-to-face, often leading to delays and red tape. | Direct digital interaction, reducing middleman involvement. |
| **Data Storage** | Physical archives/cupboards which are prone to damage. | Secure servers and cloud storage with easy backup and retrieval. |

# Describe G2C and G2B e-governance model with benefits and challenges.

## 1. G2C (Government-to-Citizen)

The **G2C** model focuses on the relationship between the government and the general public. Its primary goal is to make government services accessible to every citizen in a convenient and efficient manner.

## Key Benefits:

- ✓ **Accessibility:** Citizens can access services (like applying for a passport or birth certificate) from home without visiting a district office.


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- ✓ **Reduced Red Tape:** Digital workflows eliminate the need for middlemen, reducing delays and bureaucracy.
- ✓ **Information Dissemination:** Real-time updates on government schemes, healthcare (e.g., vaccine portals), and educational results.

## Common Challenges:

- **Digital Divide:** People in rural areas or those with low digital literacy struggle to use these platforms.
- **Language Barriers:** Many government portals are only available in a few languages, making it difficult for diverse populations.
- **Privacy Concerns:** Storing sensitive citizen data (like citizenship or biometric info) carries a high risk of data breaches.

## 2. G2B (Government-to-Business)

The **G2B** model refers to the digital interaction between government agencies and the private business sector. It aims to create a business-friendly environment by streamlining regulatory requirements.

## Key Benefits:

- **Ease of Doing Business:** Online registration of companies, patent filings, and electronic tax submissions save businesses significant time.
- **Transparency in Procurement:** E-Tendering and E-Auctioning ensure that government contracts are awarded fairly, reducing favoritism and corruption.
- **Faster Compliance:** Businesses can stay updated with new regulations and submit compliance reports instantly through digital dashboards.

## Common Challenges:

- **Interoperability:** Different government departments often use different software systems that don't "talk" to each other, making the process complex for businesses.
- **Security Risks:** Since G2B involves large financial transactions and corporate secrets, the systems are high-value targets for cyber attacks.
- **High Initial Setup Cost:** Smaller businesses may find it expensive to upgrade their internal systems to match the government's digital standards.


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# Why do we need IT infrastructure of implement in e-Governance? Explain Electronic Data Interchange.

## Why do we need IT Infrastructure for e-Governance?

- ✓ IT Infrastructure includes everything from servers and data centers to high-speed internet and security protocols. We need it for the following reasons:
- ✓ **Data Centralization & Storage:** Government operations involve massive amounts of data (citizen records, land registry, taxes). We need secure **Data Centers** and **Cloud Computing** to store and manage this information centrally.
- ✓ **Connectivity:** To bridge the gap between the government and citizens in remote areas, a strong **Network Infrastructure** (fiber optics, 4G/5G, satellite) is essential.
- ✓ **Service Delivery Platforms:** Websites and mobile apps are the "front-end" of e-Governance. Without the underlying infrastructure (servers and databases), these platforms cannot process service requests.
- ✓ **Security and Privacy:** Since government data is sensitive, the infrastructure must include **Firewalls, Encryption, and Cyber security** measures to protect against hacking and data theft.
- ✓ **Interoperability:** Infrastructure allows different government departments (e.g., Transport, Finance, Health) to share data and work together seamlessly.

## Electronic Data Interchange (EDI)

**Electronic Data Interchange (EDI)** is the computer-to-computer exchange of business documents in a **standard electronic format** between business partners or government departments.

In traditional systems, information is exchanged via paper, which is slow and prone to errors. EDI replaces this with a digital process that doesn't require human intervention.

## How EDI Works


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1. **Preparation:** Data is collected from a system (e.g., an invoice or a permit application).
2. **Translation:** The data is converted into a standard EDI format (like **ANSI X12** or **EDIFACT**) so that the receiving computer can understand it.
3. **Transmission:** The document is sent via a secure network.
4. **Receipt:** The receiving system automatically processes the document without anyone having to re-type the information.

### Key Benefits of EDI in e-Governance:

- ✓ **Speed:** Documents that took days to mail or fax now travel in seconds.
- ✓ **Accuracy:** Since there is no manual data entry, the risk of "typos" or human error is almost zero.
- ✓ **Cost Savings:** It eliminates the costs of paper, printing, storage, and postage.
- ✓ **Standardization:** It ensures that all departments and businesses use the same "language" to communicate, making the system highly efficient.

### Example in e-Governance:

A common example of EDI is **Electronic Customs Clearance**. When an importer submits cargo details, the data is sent via EDI to the Customs department, which automatically checks it against regulations and clears the goods instantly if everything matches.

## Explain the advantage of m-Governance. Describe in brief about the mobile commerce architecture.

**m-Governance** (Mobile Governance) is a sub-domain of e-Governance that leverages mobile technologies like smartphones, tablets, and wireless networks to provide services and information to citizens and businesses.

### Advantages of m-Governance

In a country where more people own mobile phones than computers, m-Governance is often more effective than traditional e-Governance.

- ✓ **Ubiquity and Reach:** It reaches people in remote or rural areas where fixed-line internet or physical offices are unavailable.


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- ✓ **Instant Access (Real-time):** Services like emergency alerts, weather updates for farmers, or transaction notifications are delivered instantly via SMS or Push notifications.
- ✓ **Location Awareness:** Mobile devices have GPS, allowing the government to provide location-based services, such as finding the nearest health center or reporting local infrastructure damage (e.g., a pothole) with precise coordinates.
- ✓ **Ease of Use:** Mobile apps are generally more user-friendly and intuitive than complex web portals, making them accessible to people with basic digital skills.
- ✓ **Cost-Effectiveness:** For the government, sending an SMS or a notification is much cheaper than printing and mailing physical notices.
- ✓ **Personalization:** Mobile devices are personal. The government can send specific alerts (like tax reminders or vaccination dates) directly to the individual concerned.

## Mobile Commerce (m-Commerce) Architecture

Mobile Commerce architecture describes the structural layers required to perform a commercial transaction (buying/selling) using a mobile device. It is a multi-tier framework.

### 1. Mobile Consumer Layer (Client)

This is the interface the user interacts with. It includes the mobile device hardware (smartphone) and the software (Mobile App or Browser). It handles the presentation of products and the user's input.

### 2. Network Layer (Communication)

This layer acts as the bridge between the mobile device and the server. It uses wireless communication technologies such as:

- ✓ **Cellular Networks:** 4G, 5G, or LTE.
- ✓ **Short-range:** Wi-Fi, Bluetooth, or NFC (Near Field Communication) for "tap-to-pay" services.

### 3. Service Gateway / Middleware Layer

This layer handles the conversion of mobile protocols into standard internet protocols. It ensures security, manages the session, and routes the request to the correct server. It acts as a translator between the mobile network and the business applications.

### 4. Application and Content Provider Layer


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This is where the actual "business logic" resides. It includes:

- ✓ **Merchant Servers:** To manage product catalogs and shopping carts.
- ✓ **Payment Gateways:** To process credit cards, digital wallets, or mobile banking API calls.
- ✓ **Database Systems:** To store user profiles, transaction history, and inventory data.

### 5. Security Layer (Cross-cutting)

Security is integrated into every level, using protocols like **SSL/TLS**, encryption, and biometric authentication (fingerprint/FaceID) to ensure that the transaction and personal data remain private.

Explain online payment processing in EPS. Why do we need security in EPS.

In the context of e-Governance and e-Commerce, an **Electronic Payment System (EPS)** is the financial backbone that allows for the paperless exchange of value.

## Online Payment Processing in EPS

The processing of an online payment involves several parties working together in a matter of seconds. The typical workflow follows these steps:

1. **Initiation:** The user (citizen or customer) selects a service and chooses a payment method (Credit Card, Digital Wallet like eSewa/Khalti, or Net Banking) on the government portal or app.
2. **Authentication:** The user enters their credentials. The system often uses **Two-Factor Authentication (2FA)**, such as an OTP (One-Time Password) sent to a mobile phone, to verify the user's identity.
3. **Authorization:** The **Payment Gateway** encrypts the data and sends a request to the user's bank (Issuing Bank). The bank checks if the user has sufficient funds or credit limit.
4. **Verification:** The bank sends an "Approved" or "Declined" message back through the payment processor to the merchant/government portal.
5. **Settlement:** Once approved, the funds are digitally transferred from the user's account to the government's or merchant's account. This usually happens in a batch at the end of the day.


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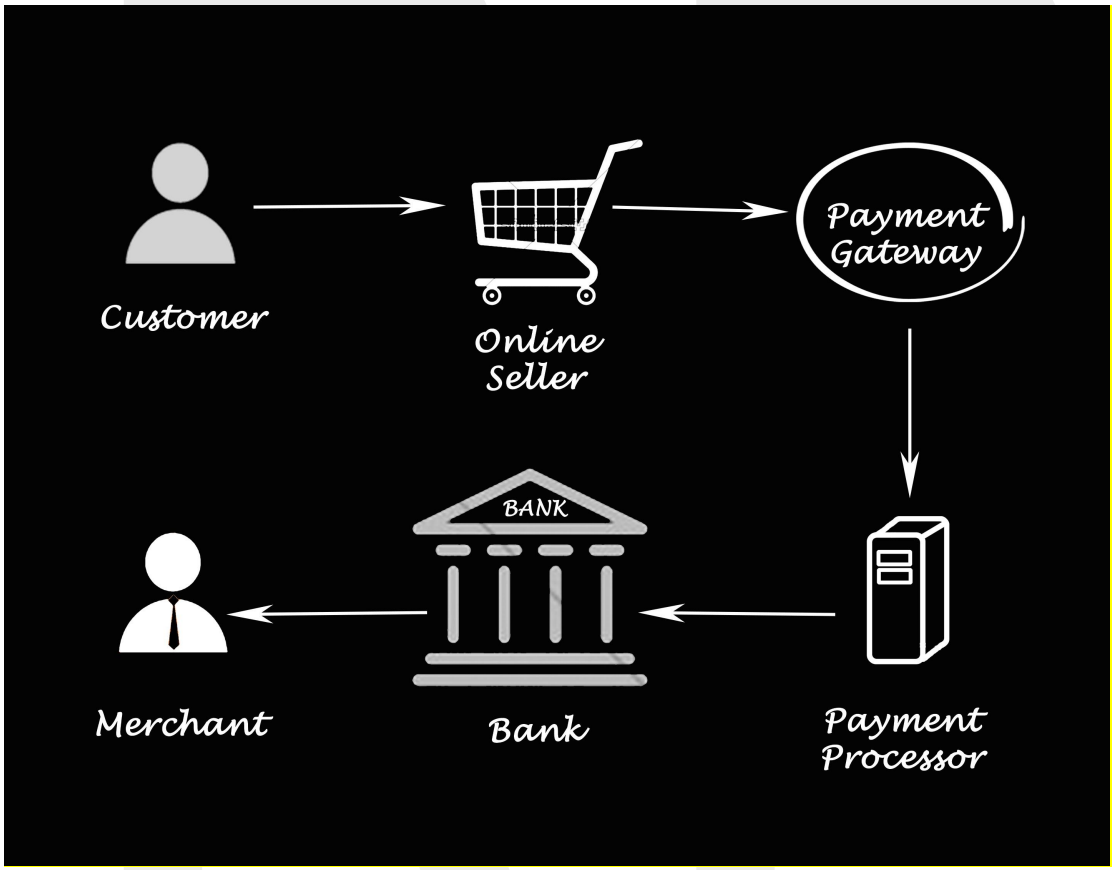
Figure: Online Payment Processing




## Why Do We Need Security in EPS?

Since EPS deals with real money and sensitive personal data, security is not just a feature—it is a mandatory requirement. We need security for the following critical reasons:

### 1. Prevention of Financial Fraud

Without security, hackers could intercept transaction details (like card numbers or CVVs) and perform unauthorized purchases. **Encryption** ensures that even if data is intercepted, it cannot be read.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### 2. Maintaining Data Integrity

Security protocols ensure that the transaction amount is not altered during transit. For example, a hacker shouldn't be able to change a $10$ payment into $1,000$ while the data is moving from the user to the bank.

### 3. Privacy and Confidentiality

Citizens trust the government with their financial history. Security measures like **SSL (Secure Sockets Layer)** prevent third parties from snooping on private financial activities.

### 4. Non-Repudiation

This ensures that a sender cannot deny having sent a payment, and a receiver cannot deny having received it. Digital signatures and timestamps provide a "digital receipt" that acts as legal proof.

### 5. Building Public Trust

If a government payment portal is hacked once, people will stop using it and return to long queues at physical offices. High security is essential to encourage the mass adoption of e-Governance services.

### 6. Protection Against Cyber Attacks

EPS systems are targets for **DDoS (Distributed Denial of Service)** attacks or **Phishing**. Robust security infrastructure (Firewalls and IDS) keeps the system "up and running" 24/7 without service interruptions.

Define biometric? Describe security issues in e-governance. How can we protect e-governance system from those issue?

## Definition of Biometrics

**Biometrics** refers to the technical term for body measurements and calculations. In IT and e-Governance, it is a security mechanism used to identify and authenticate individuals based on their unique physical or behavioral characteristics.

Common types of biometrics used in government systems (like National ID or Passport) include:

- ✓ **Physiological:** Fingerprints, facial recognition, iris scanning, and DNA.
- ✓ **Behavioral:** Voice recognition, signature dynamics, and keystroke patterns.

## Security Issues in e-Governance


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Because e-Governance platforms handle massive amounts of sensitive citizen data and financial transactions, they are prime targets for cyber threats:

- ✓ **Cyber Attacks (Malware/Ransomware):** Malicious software can freeze government databases (like land records or health data) and demand a ransom to unlock them.
- ✓ **Phishing:** Fraudsters create fake government websites to trick citizens into entering their login credentials or bank details.
- ✓ **Data Breaches & Identity Theft:** If a central database is hacked, millions of citizens' private information (Citizenship numbers, addresses, biometrics) can be stolen and sold.
- ✓ **Insider Threats:** Unauthorized access or data manipulation by government employees (intentional or accidental) remains a major risk.
- ✓ **Denial of Service (DoS/DDoS):** Overloading government servers with fake traffic to make essential services (like tax filing or results checking) unavailable to the public.
- ✓ **Interoperability Risks:** When different departments share data, a security hole in one department's "weak" system can provide a gateway to a "stronger" system.

## How to Protect e-Governance Systems

Protecting a national digital infrastructure requires a multi-layered approach:

### 1. Technical Measures

- ✓ **Encryption:** All data, whether "at rest" in a database or "in transit" over the internet, must be encrypted using strong protocols like **AES-256** or **SSL/TLS**.
- ✓ **Multi-Factor Authentication (MFA):** Moving beyond simple passwords by requiring a fingerprint scan or an OTP (One-Time Password) sent to a registered mobile number.
- ✓ **Firewalls and IDS/IPS:** Installing advanced Firewalls and Intrusion Detection Systems to monitor and block suspicious network traffic in real-time.

### 2. Administrative & Policy Measures

1. **Regular Audits:** Conducting periodic "Penetration Testing" where ethical hackers try to find holes in the government's system before real hackers do.


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2. **Disaster Recovery Plan:** Maintaining off-site data backups so that if a primary server is destroyed or hacked, the data can be restored quickly.
3. **Strict Access Control:** Implementing the "Principle of Least Privilege," where employees only have access to the specific data they need for their job.

### 3. Legal & Educational Measures

- ✓ **Cyber Laws:** Enforcing strict legal penalties for data theft and hacking to act as a deterrent.
- ✓ **Public Awareness:** Educating citizens on how to identify fake government websites and how to keep their login credentials safe.

## List out the different legal and ethical issues related to e-Governance. Write the importance of cyber law.

### 1. Ethical Issues

Ethical issues deal with what is "right" or "fair" in a digital society:

- ✓ **Digital Divide:** It is ethically questionable to provide services only online if a large portion of the population lacks internet access or digital literacy. This creates a gap between the "connected" and "unconnected."
- ✓ **Privacy vs. Surveillance:** Governments collect massive amounts of data. The ethical dilemma arises when determining how much the government should know about a citizen's private life for the sake of "security."
- ✓ **Loss of Human Touch:** Impersonal digital interfaces can make it difficult for citizens with complex or unique problems to get the empathetic assistance a human clerk might provide.
- ✓ **Data Ownership:** Who truly owns the data? Once a citizen submits their information, the government must ensure it isn't used for purposes other than what was originally intended.

### 2. Legal Issues

Legal issues focus on the framework of laws and regulations:


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- ✓ **Electronic Authentication:** Legal systems must recognize digital signatures and biometrics as being as valid as a physical handwritten signature.
- ✓ **Data Protection & Liability:** If a government database is leaked, who is legally responsible? There must be laws defining the liability of the state in protecting citizen data.
- ✓ **Jurisdiction:** In a digital world, a cybercrime might be committed in one country against the government of another. Determining which court has the power to switch or punish the criminal is a major legal hurdle.
- ✓ **Intellectual Property Rights (IPR):** Protecting the software, algorithms, and content used in e-Governance platforms from unauthorized copying or theft.

## Importance of Cyber Law

Cyber Law is the "Code of Conduct" for the internet. Without it, e-Governance would be a lawless environment where no one feels safe.

- ✓ **Legal Recognition:** Cyber law gives legal status to electronic records and digital signatures. Without this, you couldn't legally pay taxes or register a business online.
- ✓ **Prevention of Cybercrime:** It defines what constitutes a crime (hacking, identity theft, online fraud) and sets the punishments, acting as a deterrent for criminals.
- ✓ **Protection of Privacy:** It mandates how the government and private companies must handle personal data, ensuring that your private info isn't sold or misused.
- ✓ **E-Commerce Growth:** By providing a secure legal framework for online payments and contracts, cyber law builds the trust necessary for the digital economy to grow.
- ✓ **Intellectual Property Protection:** It protects the hard work of developers and creators, ensuring that software and digital innovations aren't stolen.
- ✓ **National Security:** It allows the government to legally track and prosecute those who attempt to attack a country's critical digital infrastructure (like the power grid or banking systems).


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What are the key phase of e-Governance system life cyber and how can project assessment be condicted at different phases of this life cycle?

## Key Phases of e-Governance System Life Cycle

Developing an e-Governance project is more complex than a standard software project because it involves legal, social, and political layers.

### 1. Inception & Visioning (The Planning Phase)

In this stage, the government identifies the need for a digital service (e.g., online land registration).

- ✓ **Goal:** Define the scope, identify stakeholders (G2C, G2B), and set high-level objectives.
- ✓ **Key Activity:** Creating a "Vision Document."

### 2. Analysis & Design (The Architecture Phase)

This phase involves detailed requirement gathering. Engineers study the existing "manual" process and design the "digital" workflow.

- ✓ **Goal:** Design the system architecture, database schema, and security protocols.
- ✓ **Key Activity:** Business Process Re-engineering (BPR) to simplify old manual steps.

### 3. Development & Implementation (The Build Phase)

The actual coding and hardware setup happen here.

- ✓ **Goal:** Building the software, setting up servers, and integrating payment gateways or biometrics.
- ✓ **Key Activity:** Coding, unit testing, and pilot runs in a small district or department.

### 4. Deployment & Training (The Rollout Phase)

The system is launched for the general public or the entire government body.

- ✓ **Goal:** Transitioning from the old system to the new one.
- ✓ **Key Activity:** Training government employees and launching public awareness campaigns.

### 5. Operation & Maintenance (The Sustainability Phase)


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This is the longest phase where the system is used daily.

- ✓ **Goal:** Fixing bugs, updating security patches, and ensuring 24/7 availability.
- ✓ **Key Activity:** Technical support and performance monitoring.

## Project Assessment at Different Phases

Assessment (Evaluation) is critical to ensure the project doesn't fail midway. Here is how it's conducted at each stage:

### Phase 1: Pre-Implementation Assessment (Feasibility)

- ✓ **What to assess:** Is the project technically possible? Is it financially viable?
- ✓ **Method: Cost-Benefit Analysis (CBA)** and **Readiness Assessment**. We check if the citizens have enough internet access and if the laws support digital signatures.

### Phase 2: Formative Assessment (During Design/Dev)

- ✓ **What to assess:** Is the software being built according to the requirements?
- ✓ **Method: Prototype Testing** and **User Acceptance Testing (UAT)**. Small groups of users test the interface to ensure it is "user-friendly" before the final launch.

### Phase 3: Process Assessment (During Implementation)

- ✓ **What to assess:** Is the rollout happening on time and within budget?
- ✓ **Method: Monitoring Milestones**. We track if the staff training is complete and if the hardware is installed in all target offices.

### Phase 4: Summative/Post-Implementation Assessment (After Launch)

- ✓ **What to assess:** Did the project achieve its goals? Is it reducing corruption?
- ✓ **Method: Impact Assessment** and **Citizen Satisfaction Surveys**. We measure the "Efficiency Gain" (e.g., "Earlier it took 10 days to get a license, now it takes 2 days").

### Phase 5: Technical & Security Audit

- ✓ **What to assess:** Is the system secure against new threats?
- ✓ **Method: Vulnerability Assessment and Penetration Testing (VAPT)**. Third-party experts try to hack the system to find weaknesses.

How can you manage public data in e-Governance? Describe hybrid model of e-Government.

## 1. Managing Public Data in e-Governance


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Effective data management ensures that the right information is available to the right people at the right time. It is handled through the following processes:

- **Data Collection & Digitization:** Converting old paper records (like land deeds or birth registers) into digital formats using scanning and OCR (Optical Character Recognition).
- **Centralized Databases (Data Warehousing):** Storing information in a central location so that different departments (e.g., Tax and Transport) don't have duplicate or conflicting records.
- **Data Standards & Interoperability:** Ensuring all departments use the same format (like XML or JSON) so that systems can "talk" to each other and share data seamlessly.
- **Access Control & Security:** Using **Role-Based Access Control (RBAC)** to ensure only authorized officials can edit sensitive data, while citizens can only view their own personal information.
- **Open Data Portals:** Providing non-sensitive data (like weather, traffic, or budget spending) to the public in machine-readable formats to promote transparency and innovation.
- **Data Archiving & Backup:** Regularly backing up data in off-site "Disaster Recovery Centers" to ensure that information isn't lost due to hardware failure or cyberattacks.

## 2. Hybrid Model of e-Government

The **Hybrid Model** is a strategic approach that combines different elements of governance to ensure no citizen is left behind during the digital transition. It typically refers to two main combinations:

### A. Online + Offline (Physical + Digital)

Not everyone has a smartphone or high-speed internet. A hybrid model ensures that while services are available online, **Physical Service Centers** (like *Nagari Sahayata Kendra* or Post Offices) still exist to help those who are not tech-savvy.

- **Example:** You can apply for a license online, but you go to a physical office for the biometric test and driving trial.

### B. Centralized + Decentralized Architecture


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In this technical hybrid approach, some data is kept in a massive central national server (for security and standards), while other data is managed locally by provincial or local governments (for speed and local relevance).

- **Example:** A National ID database is centralized, but local property tax records might be managed by the local municipality.

## Benefits of the Hybrid Model:

1. **Inclusivity:** It bridges the "Digital Divide" by serving both urban tech-users and rural citizens.
2. **Reliability:** If the central internet server goes down, local offline offices can still provide basic manual services.
3. **Gradual Transition:** It allows the government to slowly move away from paper without shocking the system or the elderly population who are used to traditional methods.
4. **Trust Building:** Citizens who are skeptical of "online-only" systems feel more secure knowing there is a physical office they can visit if something goes wrong with their digital application.

# Write short note

## Firewall

A **Firewall** is a network security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules. In the context of e-Governance, it acts as a digital barrier between a trusted internal network (government servers) and untrusted external networks (the internet).

## Key Functions of a Firewall:

- ✧ Packet Filtering: It inspects every "packet" of data entering or leaving the network and matches it against a set of rules. If the packet violates a rule, it is blocked.
- ✧ Access Control: It prevents unauthorized remote access to government databases while allowing legitimate citizens to access public services.


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- ✧ Threat Prevention: It protects against common cyber threats like "IP Spoofing," "DoS attacks," and "Unauthorized Login" attempts.
- ✧ Logging and Reporting: It keeps a record of all traffic and blocked attempts, which helps IT administrators identify if the system is under attack.

### Types of Firewalls:

- ✧ Packet-Filtering Firewall: The most basic type; it checks the source and destination IP addresses, protocols, and port numbers.
- ✧ Stateful Inspection Firewall: A more advanced version that tracks the "state" or context of active connections to ensure incoming traffic is part of an established, legitimate session.
- ✧ Proxy Firewall (Application Level): Acts as an intermediary between the user and the server, hiding the internal network's identity to provide extra security.
- ✧ Next-Generation Firewall (NGFW): Combines traditional firewall functions with other features like deep packet inspection (DPI) and encrypted traffic inspection.

### Importance in e-Governance:

In an e-Governance system, a firewall is the **first line of defense**. It ensures that sensitive citizen data—such as National ID details, tax records, and land ownership—remains protected from hackers and malicious software, thereby maintaining public trust in digital services.

## Extranet

In the context of e-Governance and networking, an **Extranet** is a private network that uses internet protocols and public telecommunication systems to securely share part of a government's information or operations with external stakeholders.

Think of it as an extension of an **Intranet** (which is for internal employees only) to authorized outsiders.

### Key Characteristics of an Extranet:

- ✓ **Controlled Access:** Unlike the public internet, an extranet requires authentication (username, password, or digital certificates).


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- ✓ **Targeted Users:** It is specifically designed for partners, such as private contractors, suppliers, or other government departments (G2B or G2G).
- ✓ **Security:** It uses high-level security measures like **VPNs (Virtual Private Networks)**, Firewalls, and Encryption to ensure that data moving between the government and the external partner remains private.

## Use Cases in e-Governance:

- ✓ **G2B (Government to Business):** A ministry might have an extranet where registered construction firms can submit progress reports, upload digital maps, and track payment vouchers for a public highway project.
- ✓ **Inter-Departmental Collaboration:** Different branches of government (e.g., the Department of Roads and the Nepal Electricity Authority) may use an extranet to coordinate infrastructure work so they don't dig up the same road twice.
- ✓ **Supply Chain Management:** Government procurement offices use extranets to communicate real-time inventory needs to their authorized suppliers.

# SET

**SET (Secure Electronic Transaction)** is a specialized communication protocol and a set of security standards developed in the late 1990s by **MasterCard** and **Visa**, with support from major tech companies like IBM and Microsoft.

It was designed specifically to secure credit card transactions over insecure networks, such as the internet.

## How SET Works

SET uses a combination of digital certificates and cryptography to ensure that every party in a transaction is authenticated. It involves three main components:

- ✧ The Cardholder's Digital Wallet: Software on the user's device that stores credit card information and a digital certificate.


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- The Merchant Server: The online store that receives the payment request.
- The Payment Gateway (Acquirer): The financial institution that processes the payment.

## Key Features of SET

- **Dual Signature:** This is the most unique feature of SET. It allows the customer to send encrypted information to the merchant and the bank simultaneously. The merchant sees the order details but **not** the credit card number, while the bank sees the payment details but **not** what the customer is buying.
- **Authentication:** Both the merchant and the customer must have digital certificates issued by a trusted authority. This proves the customer is the real owner of the card and the merchant is a legitimate business.
- **Data Integrity:** It ensures that the order or payment amount cannot be changed by hackers while the data is traveling through the network.

## Importance in e-Governance

In e-Governance, SET can be used for high-value government transactions (like large corporate tax payments or government procurement) where extreme security and non-repudiation are required.

## Why is it less common now?

While SET is technically superior in security, it is very complex to implement. Most modern e-commerce and e-governance systems now use **SSL/TLS (Secure Sockets Layer)** because it is easier for users and doesn't require customers to install special "wallet" software or manage complex digital certificates themselves.

# SSL

**SSL (Secure Sockets Layer)** is the standard security technology for establishing an encrypted link between a web server and a browser. This link ensures that all data passed between the web server and browsers remain private and integral.

In e-Governance, SSL is what puts the **"s"** in **HTTPS** and shows the "Padlock" icon in the address bar of a browser when you are visiting a government portal.


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## How SSL Works (The SSL Handshake)

When a citizen accesses a secure government website, the browser and the server perform an "SSL Handshake":

1. **Request:** The browser requests the server to identify itself.
2. **Certificate:** The server sends a copy of its **SSL Certificate** (which contains its public key).
3. **Validation:** The browser checks if the certificate is from a trusted CA (Certificate Authority).
4. **Session Key:** If trusted, the browser and server create a symmetric "session key" for that specific visit.
5. **Encrypted Connection:** All data (passwords, citizenship numbers, tax info) is now encrypted using this key.

## Key Functions of SSL:

- **Encryption:** It "scrambles" the data. Even if a hacker intercepts the data while you are submitting a form, they will only see a random string of characters that makes no sense.
- **Authentication:** It ensures that you are actually talking to the genuine government server and not a "spoofed" or fake website created by a phisher.
- **Data Integrity:** It ensures that the data is not altered or tampered with during transit. If even one bit of data is changed, the connection will be dropped.

## Types of SSL Certificates:

- **Domain Validated (DV):** The most basic level, verifying only who owns the domain.
- **Organization Validated (OV):** The CA checks the actual existence of the organization/government department.

**Extended Validation (EV):** The highest level of trust, requiring a thorough legal vetting process. In older browsers, this used to turn the address bar green.

## Importance in e-Governance:


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- **Public Trust:** Citizens are more likely to use digital services if they see the security padlock.
- **Compliance:** Many international standards and cyber laws make SSL mandatory for any site handling personal or financial data.
- **SEO:** Search engines like Google rank secure (HTTPS) websites higher than non-secure ones.

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